
CHAPTER 5

Discrete Probability Distributions

A number of probability mass function have proved to be useful for a large variety of practical fields in engineering, science, artificial intelligence, business and elsewhere. We will only consider the some of the most commonly used distributions, namely – *binomial* and *Poisson* and others such as *uniform*, *geometric* and *negative binomial* distributions.

5.1 Binomial Distribution

Consider a simple random experiment (or trial) whose outcome can be classified as either a **success** or **failure**. For example,

- Tossing a coin: obtaining H may be considered a success while T would represent a failure in this case.
- Results of an HIV test: testing positive for HIV is considered a success during prevalence studies such as BIAS.
- Testing a new light bulb: in quality assurance, if the bulb doesn't light when tested it is considered a success since the interest would be in the number of defectives.
- A door-to-door salesperson calls on a prospective client – the client purchases the product (success) or does not purchase (failure).

If we let $X = 1$ when the outcome is success and $X = 0$ when it is a failure, then the probability mass function of X is given by

$$p(x) = \begin{cases} p & X = 1 \\ 1 - p & X = 0 \end{cases} \quad (5.1)$$

where p , $0 < p < 1$ is the probability that the trial is a success.

Definition 5.1.1. A random variable X is said to be a *Bernoulli random variable* if its probability mass function is given by Equation (5.1) for some $p \in (0, 1)$.

Suppose that this experiment is repeated independently for n times, $n \geq 1$. We say, we have n *independent trials of the random experiment*, each of which results in success with probability p and in failure with probability $1 - p$.

Definition 5.1.2. Let X represent the number of successes that occurred from the n trials. Then X is said to be a *binomial random variable* with parameters (n, p) . The probability mass function of a binomial random variable with parameters (n, p) is given by

$$p(x) = \binom{n}{x} p^x (1 - p)^{n-x} \quad x = 0, 1, 2, \dots, n \quad (5.2)$$

Instead of writing ‘the random variable X follows a Binomial distribution with parameters (n, p) ’, we will use this neat abbreviation, $X \sim \text{Bin}(n, p)$.

To make sense of (5.2), note that for any sequence of n items, we will have x successes and $n - x$ failures and the probability of this occurring is $p^x (1 - p)^{n-x}$ due to independence of each trial. However, this would only be for one sequence of events. Note that there are $\binom{n}{x}$ different ways of obtaining x successes from a total of n trials.

It should also be noted that Equation (5.2) satisfies all the conditions of a probability mass function. In particular, the probabilities sum to 1. By binomial expansion, we have

that

$$\sum_x p(x) = \sum_{x=0}^n \binom{n}{x} p^x (1-p)^{n-x} = (p + (1-p))^n = 1.$$

Example 5.1.1. Five fair coins are flipped. If the outcomes are assumed to be independent, find the probability mass function of the number of heads obtained.

Solution 5.1.1. Let X be the number of heads (success) that appear, then $X \sim \text{Bin}(n = 5, p = 0.5)$. Hence by (5.2) we have,

$$\begin{aligned}\mathbb{P}(X = 0) &= \binom{5}{0} \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5 = \frac{1}{32} \\ \mathbb{P}(X = 1) &= \binom{5}{1} \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^4 = \frac{5}{32} \\ \mathbb{P}(X = 2) &= \binom{5}{2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^3 = \frac{10}{32} \\ \mathbb{P}(X = 3) &= \binom{5}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = \frac{10}{32} \\ \mathbb{P}(X = 4) &= \binom{5}{4} \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^1 = \frac{5}{32} \\ \mathbb{P}(X = 5) &= \binom{5}{5} \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^0 = \frac{1}{32}\end{aligned}$$

Example 5.1.2. A certain type of pill is packed in bottles of 12 pills each. 10% of the pills are chipped during the manufacturing process. Explain why the binomial distribution can provide a reasonable model for the random variable X which represent the number of chipped pills. Then find the probabilities of obtaining:

- (a) 2 chipped pills;
- (b) no chipped pills;
- (c) at least 2 chipped pills;

in a bottle.

Solution 5.1.2. The solution will be discussed in class. I'm expecting a vibrant discussions especially on the first part of the question, in which we need to justify why binomial distribution is suitable.

Expectation and Variance of a Binomial Random Variable

Let $X \sim \text{Bin}(n, p)$. Then the expected value of X is given by

$$\mathbb{E}(X) = np \quad (5.3)$$

and

$$\mathbb{V}(X) = np(1 - p) \quad (5.4)$$

A detailed proof will be provided during class as it could've been cumbersome to typeset it here. So please do ensure that you attend this class, otherwise you can check Ross's recommended book for the proof.

5.2 Poisson Distribution

Many phenomena in physics follow the Poisson probability law named in honour of the French mathematician Simeon Poisson (1781 – 1842). The classic example is the decomposition of radio-active nuclei. The following are other examples where the Poisson distribution can be used:

- In management science, the number of demands for service in a given period. (e.g. on tellers in a bank, the runways of an airport, the stock pile of a factory)
- Occurrences of accidents, errors, breakdowns and other calamities – the number

that occurs within a specified time period has a Poisson distribution under certain conditions.

Broadly, the condition for a Process process is that the events occur in time at random. Loosely, this means that an event is equally likely occur at any instant in time. That is, we only know the average time of occurrences but the exact timing of an event is random. The number of occurrences of an event during a fixed time are modeled by a Poisson distribution.

Definition 5.2.1. Suppose we are given a period of time during which events occur at random. Let λ be the average rate at which events occur per the time period. Let the random variable X be the number of events occurring during the time period. Then X has a Poisson distribution with parameter λ i.e. $X \sim \text{Poi}(\lambda)$ and has a probability mass function

$$p(x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!} & x = 0, 1, 2, \dots \\ 0 & \text{otherwise} \end{cases} \quad (5.5)$$

Example 5.2.1. Suppose you are working as a logistics manager for Unitrans and have observed that on average, the company experiences 12 break-downs per 5-day working week. You have a policy of keeping two trucks on standby. What is the probability that on any day

- (a) no standby trucks are needed?
- (b) the number of standby trucks inadequate?

Solution 5.2.1. Let X be a random variable representing the number of break-downs in a given day. It is reasonable to assume that the break-downs occur at random and that the Poisson distribution is a realistic model. Since we are interested in break-downs

per day, we also need to convert the weekly average to a daily rate. That is, 12 breakdowns per 5 days is equivalent to $12/5 = 2.4$ breakdowns per day. Hence we assume that $X \sim P(\lambda = 2.4)$, i.e.

$$\mathbb{P}(X = x) = \frac{e^{-2.4} 2.4^x}{x!} \quad x = 0, 1, 2, \dots$$

(a) No break-downs in a day means $X = 0$. Thus,

$$\mathbb{P}(X = 0) = \frac{e^{-2.4} 2.4^0}{0!} = 0.091.$$

(b) If the number of trucks on standby are inadequate, then it means that $X > 2$ and

$$\begin{aligned} \mathbb{P}(X > 2) &= 1 - \mathbb{P}(X \leq 2) \\ &= 1 - (p(0) + p(1) + p(2)) \\ &= 1 - \left(\frac{e^{-2.4} 2.4^0}{0!} + \frac{e^{-2.4} 2.4^1}{1!} + \frac{e^{-2.4} 2.4^2}{2!} \right) \\ &= 0.430 \end{aligned}$$

This implies that in only 9% of the days the company will not use standby trucks at all, but on 43% of the days they run out of standby trucks. If I were you, I'll have a talk with the COO of the company! But of course this is only, hypothetical.

Example 5.2.2. Beer cans are randomly tossed alongside A10 highway road, with an average frequency 3.2 per km.

(a) What is the probability of seeing no beer cans over a 5km stretch?

(b) What is the probability of seeing at least one beer can in 200m?

Expectation and Variance of a Poisson Random Variable

Let $X \sim \text{Poi}(\lambda)$. Then the expected value of X is given by

$$\mathbb{E}(X) = \lambda \quad (5.6)$$

and

$$\mathbb{V}(X) = \lambda \quad (5.7)$$

Proof. We are going to start with the proof of $\mathbb{E}(X)$ and make use of *Taylor expansion* of e^x given by $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$. We have

$$\begin{aligned} \mathbb{E}(X) &= \sum_{x=0}^{\infty} xp(x) \\ &= \sum_{x=0}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!} \\ &= e^{-\lambda} \sum_{x=0}^{\infty} x \frac{\lambda \cdot \lambda^{x-1}}{x(x-1)!} \\ &= \lambda e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} \end{aligned}$$

Note that we can change the variable from x to y by letting $y = x - 1$. If we do that the values of y will range from 0 to ∞ . Then we can re-write $\mathbb{E}(X)$ as:

$$\begin{aligned} \mathbb{E}(X) &= \lambda e^{-\lambda} \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} \\ &= \lambda e^{-\lambda} e^{\lambda} \\ &= \lambda \end{aligned}$$

We now turn our efforts to showing that $\mathbb{V}(X) = \lambda$. It should be noted that $\mathbb{E}(X^2) =$

$\mathbb{E}[X(X-1)] + \mathbb{E}(X)$ and

$$\begin{aligned}\mathbb{E}[X(X-1)] &= \sum_{x=0}^{\infty} x(x-1)p(x) \\ &= \sum_{x=0}^{\infty} x(x-1) \frac{e^{-\lambda} \lambda^x}{x!}\end{aligned}$$

We note that $x! = x(x-1)(x-2)!$ and also that $\lambda^x = \lambda^2 \lambda^{x-2}$. Substituting these back, we get

$$\begin{aligned}\mathbb{E}[X(X-1)] &= \sum_{x=0}^{\infty} x(x-1) \frac{e^{-\lambda} \lambda^2 \lambda^{x-2}}{x(x-1)(x-2)!} \\ &= \lambda^2 \sum_{x=2}^{\infty} \frac{e^{-\lambda} \lambda^{x-2}}{(x-2)!}\end{aligned}$$

If we let $y = x - 2$, then y will range from 0 to ∞ and we can re-write the expectation as

$$\mathbb{E}[X(X-1)] = \lambda^2 \sum_{y=0}^{\infty} \frac{e^{-\lambda} \lambda^y}{y!}$$

Note that the sum is with respect to the Poisson distribution of Y . It sums to one and thus

$$\mathbb{E}[X(X-1)] = \lambda^2.$$

Hence

$$\begin{aligned}\mathbb{V}(X) &= \mathbb{E}[X(X-1)] + \mathbb{E}(X) - [\mathbb{E}(X)]^2 \\ &= \lambda^2 + \lambda - \lambda^2 \\ &= \lambda.\end{aligned}$$

□

5.3 Other discrete distributions

5.3.1 Negative binomial distribution

The underlying scenario for the negative binomial is identified to that of the binomial distribution but with some twist. For the binomial distribution, we fixed n , number of trials and counted number of success from these n trials. But for the negative binomial distribution, we fix r , number of success and count number of failures until we have the r^{th} success. Thus, the variable X is the number of failures before obtaining r successes. We count failures, not trials!

5.3.2 Geometric distribution

Let X be the number of failures before the first success. Then X has a *geometric distribution* with parameter p (probability of a success), written, $X \sim G(p)$ and has a probability mass function given by

$$p(x) = \begin{cases} p(1-p)^x & x = 0, 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

Thus a geometric distribution is a special case of the negative binomial distribution when $r = 1$.

Example 5.3.1. Suppose that you interview job applicants in succession until you find a person that satisfies the job description. Assume that at each interview, the probability of finding the right person is 0.3.

- (a) What is the probability that you appoint the third person you interview?
- (b) What is the probability that you will need to do five or more interviews?